



HOW TO USE THIS SET

20 questions, 5 options each (A-E), mixed difficulty, 13 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

How does an enzyme speed up a biochemical reaction?

- A. By raising the activation energy
- B. By lowering the activation energy
- C. By being used up as the reaction proceeds
- D. By changing the nature of the end product
- E. By acting on any substrate it meets

2.

EASY

Pepsinogen is secreted by the stomach as an inactive zymogen. What converts it to active pepsin?

- A. Trypsin from the pancreas
- B. Bile from the gallbladder
- C. HCl or active pepsin
- D. Secretin from the duodenum
- E. Rennin in the gastric juice

3.

EASY

Which was the first gastrointestinal hormone to be discovered?

- A. Gastrin
- B. Cholecystokinin
- C. Secretin
- D. Somatostatin
- E. Ghrelin

4.

EASY

Ghrelin is released by endocrine cells of the stomach during hunger. It stimulates feeding by acting on the:

- A. Anterior pituitary
- B. Pancreatic islets
- C. Gallbladder
- D. Duodenal mucosa
- E. Hypothalamus

5.

MEDIUM

CASE

A 24-year-old woman gets watery diarrhoea within two hours of drinking milk. The symptoms stop when she avoids dairy.

Which mechanism best explains the diarrhoea?

- A. Undigested lactose raises osmotic pressure and draws water into the large intestine
- B. Intestinal bacteria ferment the lactose and release CO₂ gas into the bowel
- C. Loss of pancreatic lipase leaves dietary fat undigested in the stool
- D. Excess gastrin increases the secretion of gastric juice after meals
- E. Lack of somatostatin speeds up gastric motility after every meal

6.

MEDIUM

CASE

A 3-month-old breast-fed infant digests the milk protein casein efficiently.

Which enzyme does the lecture assign to casein digestion in infants?

- A. Pepsin
- B. Rennin
- C. Lactase
- D. Trypsin
- E. Chymotrypsin

7.

MEDIUM

An enzyme hydrolyses the peptide bond of the N-terminal amino acid of a peptide. How is it classified?

- A. Endopeptidase
- B. Carboxypeptidase
- C. Aminopeptidase
- D. Ribozyme
- E. Zymogen

8.

MEDIUM

CASE

A purified pancreatic enzyme splits a test peptide in two at an internal bond. The carbonyl group of that bond is contributed by lysine.

Which enzyme is it?

- A. Chymotrypsin
- B. Elastase
- C. Carboxypeptidase A
- D. Trypsin
- E. Carboxypeptidase B

9.

MEDIUM

CASE

A 50-year-old man with chronic pancreatitis has weight loss and bulky, greasy stools.

Digestion of which nutrient is affected most, and why?

- A. Starch, because amylase is produced only by the pancreas
- B. Fat, because the pancreas is the main source of lipase
- C. Lactose, because lactase is a pancreatic enzyme
- D. Casein, because rennin is a pancreatic enzyme
- E. Protein, because pepsin is a pancreatic enzyme

10.

MEDIUM

CASE

A tumour of G cells releases its hormone continuously, even between meals.

Which set of effects is expected?

- A. More HCl and pepsin, faster gastric motility, epithelial proliferation
- B. Less gastric acid secretion, with strong contraction of the gallbladder
- C. Less insulin and glucagon, with less secretion of pancreatic juice
- D. More feeding, through a direct action on the hypothalamus
- E. Slower gastric motility, with more secretion of bile

11.

MEDIUM

CASE

Thirty minutes after a fatty meal, an ultrasound shows the gallbladder contracting strongly and emptying bile into the small intestine.

Which hormone is responsible, and where is it secreted?

- A. Secretin — duodenum and upper jejunum
- B. Gastrin — G cells of the stomach
- C. Somatostatin — δ cells of the stomach
- D. Cholecystokinin — duodenum and jejunum
- E. Ghrelin — endocrine cells of the stomach

12.

MEDIUM

Acidic gastric juice empties into the duodenum and triggers release of a hormone. By which mechanism does this hormone reduce gastric acid secretion?

- A. It inhibits gastrin release from G cells
- B. It inhibits pepsin secretion by the stomach
- C. It contracts the gallbladder to release bile
- D. It blocks insulin and glucagon secretion
- E. It acts on the hypothalamus to stop feeding

13.

MEDIUM

CASE

A patient receives an infusion of somatostatin, the "GH release inhibitory factor".

Secretion of which pair of anterior pituitary hormones is expected to fall?

- A. ACTH and prolactin
- B. LH and FSH
- C. Insulin and glucagon
- D. Gastrin and CCK
- E. GH and TSH

14.

MEDIUM

A catalytic molecule works by transesterification and phosphodiester bond hydrolysis. It is inactivated by heating above its melting temperature or by changing essential nucleotides. What is it?

- A. A globular protein enzyme
- B. A zymogen
- C. A peptide hormone
- D. A ribozyme
- E. A lipopeptide

15.

HARD

CASE

An infant is born without enteropeptidase. Pepsin activity in the stomach is normal.

According to the lecture's activation figure, what happens to protein digestion in the small intestine?

- A. Only trypsin is missing; the other pancreatic proteases work normally
- B. Only the carboxypeptidases are missing; trypsin still works normally
- C. Nothing changes, because HCl activates the pancreatic zymogens
- D. Only elastase is missing, so bonds next to Ala, Gly and Ser survive
- E. Trypsin is not formed, so the other pancreatic zymogens stay inactive

16.

HARD

CASE

Peptide 1 ends in lysine at its C-terminal end; peptide 2 ends in valine. Enzyme X releases the terminal lysine from peptide 1 but leaves peptide 2 intact. Enzyme Y does the opposite.

Identify X and Y.

- A. X = carboxypeptidase B; Y = carboxypeptidase A
- B. X = carboxypeptidase A; Y = carboxypeptidase B
- C. X = trypsin; Y = chymotrypsin
- D. X = aminopeptidase; Y = elastase
- E. X = trypsin; Y = carboxypeptidase A

17.

HARD

CASE

A meal rich in fat and protein leaves the stomach as acidic chyme and enters the duodenum.

Which pair of responses does the lecture predict?

- A. Gastrin speeds gastric motility; ghrelin promotes further feeding
- B. Secretin contracts the gallbladder; CCK inhibits gastrin release
- C. GIP raises gastric acid secretion; somatostatin raises pancreatic juice
- D. Secretin inhibits gastrin release; CCK contracts the gallbladder
- E. CCK stimulates pepsin secretion; secretin acts on the hypothalamus

18.

HARD

CASE

After a 16-hour fast, a volunteer feels intensely hungry. A stomach-derived hormone that acts on the hypothalamus is raised.

Which description of this hormone is correct?

- A. A 14-amino-acid peptide from δ cells that inhibits growth hormone release
- B. A 28-amino-acid peptide from δ cells that inhibits insulin and glucagon
- C. A 28-amino-acid lipopeptide with an 8-carbon fatty acid that counteracts leptin
- D. A peptide from G cells that is released by amino acids in the gastric lumen
- E. A peptide from the duodenum and jejunum that inhibits gastric motility

19.

HARD

CASE

A synthetic peptide has the sequence Asp-Thr-Lys-His-Glu-Ser-Asn-Gln-Thr.

Using the cleavage specificities given in the lecture, which two enzymes together would cut it into three fragments?

- A. Trypsin and chymotrypsin
- B. Chymotrypsin and elastase
- C. Carboxypeptidase A and carboxypeptidase B
- D. Carboxypeptidase B and elastase
- E. Trypsin and elastase

20.

HARD

CASE

A patient has the whole pancreas removed and receives no enzyme replacement.

Which statement about the remaining digestive enzyme activity fits the lecture?

- A. No amylase remains, because amylase is purely pancreatic
- B. Pepsin still digests protein, but trypsin and chymotrypsin are lost
- C. No lipase remains, because mouth and stomach produce none
- D. Rennin is lost, because rennin is a pancreatic protease
- E. Carboxypeptidases remain, because the stomach secretes them

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	By lowering the activation energy — Enzymes lower activation energy. They are not used up, do not alter the end product, and are highly specific.
2	C	HCl or active pepsin — Pepsinogen is activated by HCl or autocatalytically by pepsin molecules that are already active.
3	C	Secretin — The lecture names secretin as the first GI hormone discovered.
4	E	Hypothalamus — Ghrelin acts on the hypothalamus to stimulate feeding, counteracting leptin.
5	A	Undigested lactose raises osmotic pressure and draws water into the large intestine — Lactose in the gut raises osmotic pressure, pulling water from tissue into the large intestine. Fermentation to CO ₂ explains the cramps, not the diarrhoea.
6	B	Rennin — Rennin digests milk protein (casein) in infants. Lactase is in the lecture's enzyme list but acts on lactose, the milk sugar.
7	C	Aminopeptidase — Exopeptidases work at the ends: aminopeptidases at the N-terminal amino acid, carboxypeptidases at the C-terminal one. Endopeptidases cut inner bonds.
8	D	Trypsin — Trypsin cleaves only where the carbonyl group comes from Arg or Lys. Carboxypeptidase B also handles Arg and Lys, but at the C-terminal end.
9	B	Fat, because the pancreas is the main source of lipase — EPI is a deficiency of amylase, protease and lipase, with fats affected most. Mouth and stomach make only small amounts of lipase; the pancreas makes the larger amount.
10	A	More HCl and pepsin, faster gastric motility, epithelial proliferation — G cells release gastrin, which increases gastric juice (HCl + pepsin), increases gastric motility and drives proliferation of gastric epithelium.
11	D	Cholecystikinin — duodenum and jejunum — CCK is secreted by the duodenum and jejunum in response to fat and protein products, and strongly contracts the gallbladder.
12	A	It inhibits gastrin release from G cells — The hormone is secretin. It reduces acid by inhibiting gastrin release from G cells — and it stimulates, rather than inhibits, pepsin secretion.
13	E	GH and TSH — Somatostatin inhibits GH and TSH from the anterior pituitary. It also inhibits insulin, glucagon, gastrin and CCK, but none of those is a pituitary hormone.
14	D	A ribozyme — A ribozyme is an RNA with catalytic activity; these are its mechanism and its two listed ways of being inactivated.
15	E	Trypsin is not formed, so the other pancreatic zymogens stay inactive — Enteropeptidase converts trypsinogen to trypsin; trypsin then activates chymotrypsinogen, proelastase and the procarboxypeptidases.
16	A	X = carboxypeptidase B; Y = carboxypeptidase A — Carboxypeptidase B removes the basic amino acids Arg and Lys; carboxypeptidase A removes Ala, Ile, Leu and Val. Both work at the C-terminal end.
17	D	Secretin inhibits gastrin release; CCK contracts the gallbladder — Acid in the duodenum releases secretin, which inhibits gastrin release. Fatty acids and amino acids release CCK, which contracts the gallbladder.
18	C	A 28-amino-acid lipopeptide with an 8-carbon fatty acid that counteracts leptin — This is ghrelin: a lipopeptide of 28 amino acids with a covalently attached 8-carbon fatty acid. Somatostatin also has a 28-amino-acid form but comes from δ cells and inhibits.
19	E	Trypsin and elastase — Trypsin cuts at Lys and elastase cuts at Ser — two internal cuts, three fragments. There is no Trp, Tyr, Phe, Met or Leu for chymotrypsin, and the C-terminal Thr suits neither carboxypeptidase.
20	B	Pepsin still digests protein, but trypsin and chymotrypsin are lost — Pepsin is gastric, so it survives; trypsin, chymotrypsin and the carboxypeptidases are pancreatic. Salivary amylase and small amounts of mouth and stomach lipase also remain.